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Batch :- 1	Class :- SY.BSC.IT
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PRACTICAL 25

Aim:- Handle missing values in a dataset. Replace missing values with the column mean.

Input:-

```

1  object d25 extends App {
2
3      val sales:List[java.lang.Double] = List(100.0, 200.0, null, 300.0, null)
4
5      val v = sales.filter(_ != null)
6
7      val mean = v.map(_.toDouble).sum / v.length
8
9      println("Mean Sales: " + mean)
10
11     val n = sales.map { value =>
12         if (value == null) mean
13         else value
14     }
15     println("Original Data:")
16     println(sales.mkString(", "))
17
18     println("After Replacing Missing Values:")
19     println(n.mkString(", "))
20 }
21

```

Output:-

```

Run  d25 x
C:\Program Files\Java\jdk1.8.0_181\bin\java.exe ...
Mean Sales: 200.0
Original Data:
100.0, 200.0, null, 300.0, null
After Replacing Missing Values:
100.0, 200.0, 200.0, 300.0, 200.0
Process finished with exit code 0

```

Aim:- Perform logistic regression using Breeze. Classify a dataset with binary labels.

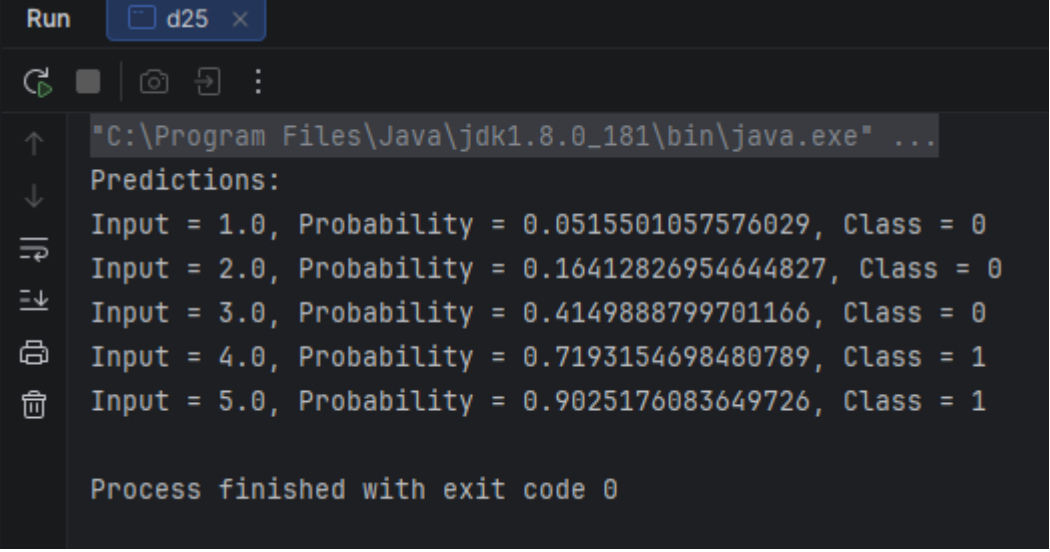
Input:-

```
build.sbt × d25.scala
1 ThisBuild / scalaVersion := "2.12.18"
2
3 libraryDependencies += "org.scalanlp" %% "breeze" % "2.1.0"
4
5 lazy val root = (project in file("."))
6   .settings(
7     name := "d25"
8   )
9
```

```
build.sbt d25.scala ×
1 import breeze.linalg._
2
3 object d25 extends App {
4
5   val x = DenseVector(1.0, 2.0, 3.0, 4.0, 5.0) // Dataset
6   val y = DenseVector(0.0, 0.0, 0.0, 1.0, 1.0) // Binary labels
7
8   // Initial weight and bias
9   var weight = 0.0
10  var bias = 0.0
11
12  // Sigmoid function
13  def sigmoid(z: Double): Double = {
14    1.0 / (1.0 + math.exp(-z))
15  }
16
17  // Train the model
18  for (i <- 1 to 1000) {
19    for (j <- 0 until x.length) {
20      val prediction = sigmoid(weight * x(j) + bias)
21      val error = prediction - y(j)
22      weight = weight - 0.01 * error * x(j)
23      bias = bias - 0.01 * error
24    }
25  }
```

```
26
27  // Classify the data
28  println("Predictions:")
29  for (i <- 0 until x.length) {
30    val probability = sigmoid(weight * x(i) + bias)
31    val result = if (probability >= 0.5) 1 else 0
32    println("Input = " + x(i) + ", Probability = " + probability + ", Class = " + result)
33  }
34 }
35
```

Output:-



The screenshot shows a Java IDE's Run console. At the top, there's a 'Run' button and a tab labeled 'd25'. Below the toolbar, the command prompt shows the execution of 'java.exe'. The output displays a list of predictions for five different input values, each with a calculated probability and a predicted class (0 or 1). The process concludes with an exit code of 0.

```
"C:\Program Files\Java\jdk1.8.0_181\bin\java.exe" ...  
Predictions:  
Input = 1.0, Probability = 0.0515501057576029, Class = 0  
Input = 2.0, Probability = 0.16412826954644827, Class = 0  
Input = 3.0, Probability = 0.4149888799701166, Class = 0  
Input = 4.0, Probability = 0.7193154698480789, Class = 1  
Input = 5.0, Probability = 0.9025176083649726, Class = 1  
  
Process finished with exit code 0
```